

ActPlane: Letting Agents Harness Their Own System-Level Behavior with eBPF

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Abstract

AI coding agents require harnesses to apply behavioral policies for effective and safe operation: e.g. do not leak secrets, run tests before committing. Our empirical study of 64 projects shows that while project harness instructions are declared in natural language, 64% consist of behavioral policies.

Yet existing agent harnesses fail to connect intent-level declarations to deterministic system-level policies, leaving a *semantic gap* and making it difficult to ensure compliance or follow rules. While 83% of directives involve system-level behavior, prompt constraints operate at the probabilistic intent level. 81% of projects require cross-event state tracking, but tool-call guardrails lose track of system-level actions. 74% of system-level rules require intent-level project or task context to evaluate, while current OS-level mechanisms expect predefined static policy and return only system events without semantic context.

To address this, we argue the policy engine should be exposed as a programmable interface, allowing agents to dynamically define and adapt their policy for their own system-level actions based on their intent context. We realize this in ActPlane, which allows agents to declare cross-event behavior as labeled information-flow control (IFC) policies under a layered policy model, observing and enforcing the agents' actions across process, file, and network boundaries with semantic feedback. We implement ActPlane with eBPF and evaluate on 607 policies from the empirical study, showing that ActPlane improves policy compliance by XX% over existing enforcement mechanisms and improves repository-grounded checklist satisfaction by XX pp, with low end-to-end overhead.

CCS Concepts: • **Computer systems organization** → *Embedded and cyber-physical systems*; • **Security and privacy** → *Software and application security*; • **Software and its engineering** → *Software safety*.

Keywords: AI agents, eBPF, BPF-LSM, labeled information-flow control, information-flow control, provenance, semantic feedback

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1 Introduction

AI coding agents have become widely deployed tools for software development [13]. They operate as long-running processes with access to shells, source trees, package managers, and external APIs, autonomously writing code, running tests, and managing infrastructure across extended sessions.

As agents gain autonomy, they require harnesses to apply behavioral policies. Projects encode such policies in instruction files such as `CLAUDE.md` and `AGENTS.md` [5, 17]. Our empirical study of 64 projects extracts 2,116 statements and finds that 64% are behavioral directives, 83% of which involve system-level behavior, and 81% of projects require cross-event state tracking (§3). A directive is *per-event* if it can be checked against a single operation (“do not run `git push -force`”) and *cross-event* if it requires state across operations (“run tests before committing”—the commit is forbidden unless a test execution has been observed since the last source change). Because LLM compliance is inherently probabilistic [16], policy cannot reside inside the model [3, 26]: even cooperative agents routinely violate declared constraints through long-horizon planning errors, shell-script side effects, and opaque tool behavior [?].

Enforcing these policies faces two compounding gaps. First, a *context gap*: 74% of system-level directives are not self-contained—they reference project domains (“run the test suite”) or delegate judgment to the current task (“unless explicitly requested”) that cannot be resolved without the agent’s own project and session context (§3). A static policy authored by an external administrator cannot supply this context.

Second, a *semantic gap* between intent and system actions. Intent-level guards [23, 27] lose coverage when agents shell out or spawn subprocesses: the same `git push` can be reached through a tool call, a `bash -c` string, or a compiled binary. OS-level policy [6, 15, 19] is unbypassable but requires policies to be statically pre-written and returns opaque errors with no connection to declared intent. The agent—the entity that holds both the project context and the task intent needed to instantiate 74% of its own rules—has no programmatic interface to define or evolve the constraints governing its system-level actions.

Our insight is that the policy engine should be exposed as a programmable interface that lets agents become the control plane of their own system-level actions. Because the agent is the only entity that holds the project structure and current task intent needed to concretize its rules, the agent

must author its own enforcement policy—trusted to honestly declare intent, but not trusted to reliably follow it during execution.

ActPlane is such an interface. Agents and developers declare behavioral policies in a compact DSL that agents themselves can generate from natural-language directives; ActPlane compiles them into labeled information-flow rules and loads them into an eBPF/BPF-LSM backend. Named labels propagate across process, file, and network objects at kernel hooks, encoding execution history as checkable state: when a process reads a file labeled SECRET, the process acquires that label; when it forks, the child inherits it. When a labeled subject matches a rule, ActPlane returns semantic feedback—which rule was matched, why, and what alternative path satisfies it. Each rule independently selects notify (observe), block (pre-operation denial), or kill (post-operation termination). Policies are layered: higher-authority rules (e.g., platform-level “do not leak secrets”) cannot be weakened by the agent, while agent-authored rules can be revised as task intent evolves.

This paper makes three contributions.

1. An **empirical study** of instruction files from 64 popular projects, showing that cross-event information-flow policies are pervasive (81% of repositories) and that 74% of system-level directives require project or task context that only the agent can supply.
2. **ActPlane**, a programmable OS-level control plane for agent harnesses. ActPlane compiles behavioral policies into eBPF/BPF-LSM labeled information-flow rules, observes and enforces them across process, file, and network boundaries, returns semantic feedback that reconnects rule matches to intent-level remediation, and supports layered policies where higher-authority constraints cannot be weakened by the agent.
3. An **evaluation** on 607 OS-level directives from the empirical study, measuring DSL expressiveness, LLM-based natural-language-to-DSL translation correctness, system correctness across bypass paths, per-event overhead, and feedback effectiveness on repository-grounded OctoBench tasks.

2 Background and Motivation

2.1 Agent Harnesses and Behavioral Policies

An agent harness is the non-model infrastructure that mediates between an LLM agent and its execution environment [3, 26]. Harness components include tool dispatch, memory management, orchestration, observation, enforcement, and feedback. This paper focuses on the observation, enforcement, and feedback components: the mechanisms that ensure an agent’s behavioral policies hold at runtime.

Behavioral policies are constraints that govern an agent’s observable effects rather than its internal reasoning. Projects encode them in instruction files such as CLAUDE.md and

AGENTS.md [5, 17]: do not push directly to main; run tests before committing; never expose secrets to the network. These policies differ from coding-style directives (e.g., “prefer const over let”) in that they produce observable effects at the system-call boundary.

2.2 The Semantic Gap

An agent’s execution spans three abstraction levels.

Intent is the agent’s own goals and self-constraints, i.e., what it declares it will and will not do. **Action** is the tool calls the agent issues through its runtime (`read_file`, `run_command`, `edit_file`), mediated by the agent framework.

Behavior is the actual OS-level operations that result: open, `execve`, connect, and fork.

The mapping from action to behavior is many-to-many and opaque. A single `run_command("make")` triggers hundreds of system calls. The same `git push` can be reached through a direct tool call, a `bash -c` string, a Python subprocess, or a compiled helper binary. This opacity creates a *semantic gap*: policies declared at the intent level cannot be reliably enforced at a single intermediate level.

Enforcement placed at the intent level (prompt instructions) is probabilistic: LLM compliance degrades with context length and task complexity [16]. Enforcement placed at the action level (tool-call guards [23, 27]) is deterministic but bypassable: it intercepts tool calls, not the OS behavior they produce. Enforcement placed at the behavior level (syscall filters, sandboxes [6, 19]) is complete but returns opaque `-EPERM` failures with no connection to the agent’s declared policies. Each level addresses a different part of the stack; none spans the full gap from intent to behavior.

A cooperative agent does not intentionally bypass tool-layer guards, but the opacity of the action-to-behavior mapping causes unintentional bypass in routine operation. Three representative scenarios illustrate this. (1) An agent calls `run_command("npm test")`; the test harness internally invokes `git push` to a staging server. The tool-call guard sees `npm test` (a permitted action), while the kernel observes the forbidden `git push` behavior. (2) An agent calls `read_file(".env")` to inspect a configuration value, then later calls `run_command("curl api.example.com")`. Each tool call is individually permitted, but the combination constitutes secret exfiltration, a cross-event condition visible only to an enforcer that tracks information flow from file read to network connect. (3) An agent writes a Python script containing `subprocess.run(["git", "push", "-force"])` and executes it. The tool-call guard sees `run_command("python script.py")`; the kernel observes the nested `execve("git", ["push", "-force"])`.

2.3 Threat Model

ActPlane targets a *cooperative but probabilistic* agent: trusted to honestly declare its intent, but not trusted to reliably follow that intent during execution. The agent is not adversarial—it does not attempt to defeat the enforcement mechanism

through obfuscation or exploitation. However, because LLM inference is inherently stochastic, the agent may trigger any given policy rule with non-zero probability at each step, regardless of context length or prompt design. Over a long session with many steps, the cumulative probability of at least one rule match approaches certainty. This matches real coding-agent failures: agents intend to obey instructions but deviate through long-horizon planning errors, prompt injection, shell-script side effects, or opaque tool behavior.

Trust boundary. The eBPF programs and BPF maps that propagate labels and check rules run in kernel space and cannot be modified by the agent. The policy DSL compiler and feedback formatter run in a trusted userspace process (`actplane run`) that is not part of the agent’s process tree. The monitored domain is the agent’s entire process tree, including all descendants spawned through tools, shells, or subprocesses.

Multi-layer policy. Policies are layered by authority. A platform or operator defines root-level constraints (e.g., “do not leak secrets,” “do not `rm -rf /`”) that lower layers cannot weaken or bypass. The agent may author and revise its own rules within these bounds as task intent evolves; runtime policy revisions are explicit transactions that cannot retroactively legalize past violations. Untrusted or third-party sub-agents execute as subprocesses in a lower layer, subject to both their parent’s rules and any additional constraints—reducing to a conventional sandbox for fully untrusted code.

Out of scope. ActPlane does not defend against an agent that deliberately exploits kernel vulnerabilities or manipulates the eBPF loader. Such attacks require `CAP_BPF` or root privileges that the agent does not hold. Side-channel inference (deducing policy rules from `-EPERM` patterns), deliberate label pollution (triggering over-tainting to cause denial of service), and feedback-oracle attacks (using remediation strings to restructure an approach toward a forbidden goal through compliant steps) all require adversarial intent and are outside this threat model.

3 Empirical Study

This section characterizes behavioral policies in real agent instruction files and identifies the subset that requires cross-event information-flow observation and enforcement.

3.1 Methodology

We collected `CLAUDE.md` and `AGENTS.md` files from 64 popular open-source repositories, extracted 2,116 statements, and classified each along four axes: content type (description vs. directive), topic (12 categories), enforcement level (semantic-only, content, per-event, cross-event), and context requirement (none, project, task). Enforcement level is assigned by a waterfall decision procedure: each directive exits at the first level whose predicate it satisfies.

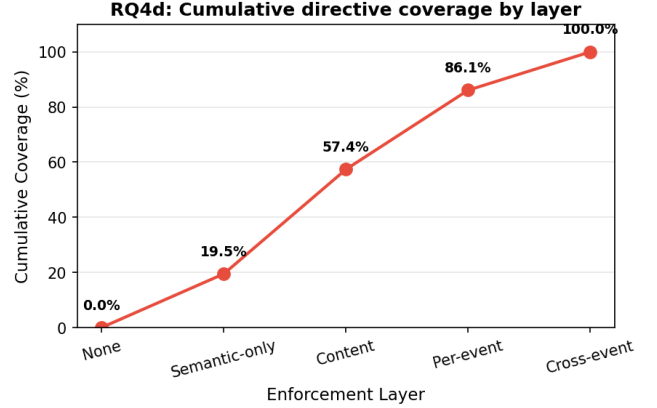


Figure 1. Cumulative directive coverage by enforcement layer. Linters cover 55.4%; per-event matching reaches 84.2%; cross-event IFC closes the remaining 15.8%.

3.2 Findings

Instruction files are behavioral policies. 64% of statements are directives; 83% of directives involve system-observable behavior (commands, files, network). We define a *per-event* directive as one whose predicate can be evaluated by inspecting a single system call (e.g., “do not run `git push -force`”), and a *cross-event* directive as one whose predicate requires state accumulated across multiple operations (e.g., “run tests before committing” requires observing a test execution after the most recent source change).

Enforcement is layered. Content inspection covers 38.2% of directives, per-event syscall matching covers 28.8%, and 15.8% require cross-event state tracking (Figure 1).

Cross-event policies are pervasive at the project level. Although cross-event directives are 15.8% of all directives, 81% of repositories contain at least one, and 43% require all four enforcement layers (Figure 2).

Most system-level directives are not self-contained. Of the 1,127 system-level directives (content + per-event + cross-event), only 26% are fully self-contained: the directive text specifies all patterns, paths, and commands needed to instantiate a concrete enforcement rule. 64% require *project context*—the directive references a domain (“run the test suite,” “use the project’s linter”) whose concrete realization must be resolved from the repository. 9% require *task context*—the directive delegates a judgment to the current session (“unless explicitly requested,” “keep changes focused”). This quantifies the gap between a natural-language policy and a deployable enforcement rule, and motivates a policy architecture in which the agent—the entity that holds both project structure and task intent—authors its own rules.

3.3 Implications

Per-event policies can be enforced by matching a single system call. Cross-event policies cannot: they require tracking

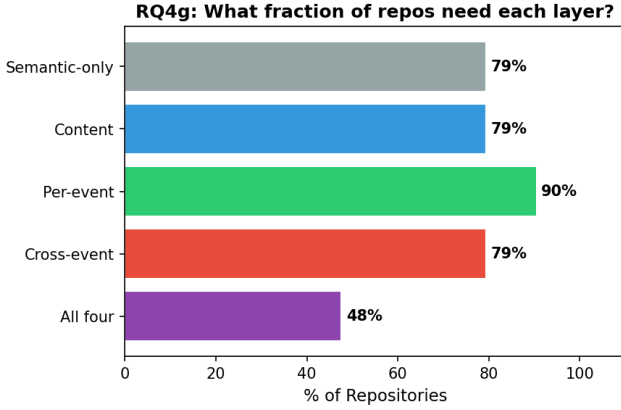


Figure 2. Fraction of repositories requiring each enforcement layer. 81% need cross-event tracking; 43% need all four layers.

state across multiple operations. Enforcing them requires a mechanism that (1) propagates provenance across process, file, and network boundaries, (2) encodes it as checkable state, and (3) connects rule matches back to declared intent for semantic feedback. Labeled information-flow control satisfies all three requirements.

The context-requirement finding adds a fourth constraint: (4) the mechanism must accept policies authored by the agent itself, because 74% of system-level rules cannot be instantiated without project or task context that only the agent possesses. A static, administrator-authored policy cannot close this gap.

4 Design

ActPlane is a programmable OS-level control plane for agent harnesses. It compiles behavioral policies into labeled information-flow rules at OS kernel hooks, observing and enforcing agent behavior and returning semantic feedback when a rule is matched. Figure 3 summarizes the pipeline.

4.1 Policy Model

ActPlane policies are object-level information-flow rules over three node types: processes (identified by pid and command name), files (keyed by path hash), and endpoints (keyed by IPv4 address and mask). The model operates at object granularity rather than byte granularity [14], trading precision for low-overhead whole-system coverage.

4.1.1 Labels and Propagation. Each node carries a 64-bit label bitmask. A *source* declaration introduces labels: an *exec* source labels processes that execute a matching binary, a *file* source labels files matching a path pattern, and an *endpoint* source labels network endpoints matching an address range. Let $L(x)$ denote the label bitmask of node x . Labels propagate through fixed transfer functions:

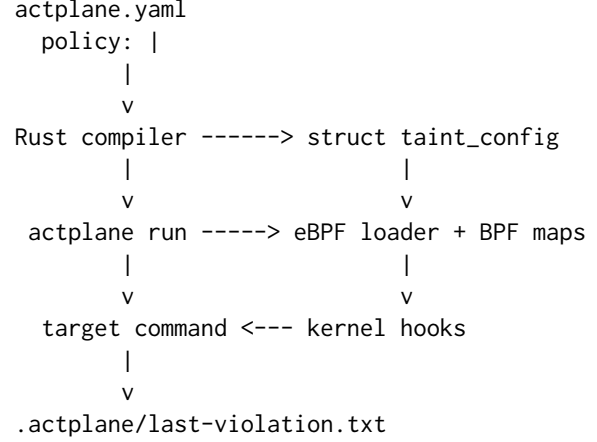


Figure 3. ActPlane pipeline. The Rust compiler lowers the policy DSL into a fixed-size binary configuration. The eBPF loader writes it into read-only data, attaches to kernel hooks, and reports rule matches as NDJSON. The runner maps rule identifiers back to human-readable feedback.

- **fork**($p \rightarrow c$): $L(c) := L(c) \cup L(p)$.
- **exec**(p, f): $L(p) := L(p) \cup L(f) \cup S(f)$, where $S(f)$ is the set of source labels matching f .
- **read**(p, f): $L(p) := L(p) \cup L(f)$.
- **write**(p, f): $L(f) := L(f) \cup L(p)$.
- **connect**(p, e): $L(e) := L(e) \cup L(p)$.

All transfer functions are monotonic: labels accumulate and are never removed by propagation. The only mechanism that removes a label is an explicit *declassify* directive in the DSL, which clears specified labels when the agent executes a designated gate binary (e.g., a redaction tool). This monotonicity bounds over-tainting: a node's label set grows at most to the union of all source labels reachable through observed flow edges.

These transfer functions maintain a provenance invariant: if a process p carries label ℓ , then information from some source tagged with ℓ has reached p through observed OS-level flow edges.

4.1.2 Rules and Effects. Unlike traditional taint analysis and IFC systems that prioritize completeness (minimizing false negatives, tolerating high false positives) [7, 11], ActPlane targets *precision*: false positives that block legitimate operations disrupt the agent's development workflow and erode trust in the harness. The design therefore favors object-granularity labels with explicit declassification gates over byte-level tracking, accepting bounded under-tainting in exchange for near-zero false positives on compliant executions.

A rule contains one or more clauses. Each clause starts with an action verb (notify, block, or kill) followed by an operation (exec, read, write, unlink, connect, or open), a target pattern, optional positional arguments, a Boolean

```

source AGENT = exec "claude"
source SECRET = file "**/*.env"

# Per-event: from CoplayDev/unity-mcp
rule no-push-main:
  kill exec "git" "push" "main" if AGENT
  because "Do not push to main; open a PR instead."

# Cross-event: from chenhg5/cc-connect
rule tests-before-commit:
  block exec "git" "commit"
  if AGENT unless after exec "go" since write "**/*.go"
  because "Run `go test ./...` before committing."

```

Figure 4. Two ActPlane rules drawn from real projects. The first is a per-event rule (single `execve` match); the second is a cross-event rule requiring temporal state (after. . . since).

expression over required and forbidden labels, and an optional condition (target allow-list, lineage gate, or temporal after-gate).

Each clause declares its own effect that determines how the kernel responds to a match.

Notify: the kernel emits a match event and allows the operation to proceed.

Block: a BPF-LSM hook returns `-EPERM` before the operation commits. Block is available only when the bpf LSM is active in the running kernel; it is not silently downgraded.

Kill: the kernel sends `SIGKILL` to the matching task. From a BPF-LSM hook, kill also returns `-EPERM`; from a tracepoint, kill is post-operation termination.

When multiple rules match the same event, the strongest effect applies in the order `kill > block > notify`.

4.1.3 Policy Language Scope. The DSL is not a general-purpose mandatory access control policy. Its primitives map to recurring agent workflow patterns: “after tests,” “through a review tool,” “data derived from untrusted input,” and “read-only sub-agent.” The grammar is small so that policy authors can translate natural-language instructions into source/sink/gate shapes without security-policy expertise.

The intermediate DSL is a deliberate design choice. The IFC engine’s backend configuration (`struct taint_config`) consists of hundreds of fixed-size rule entries, bitmask allocations, and match-kind encodings across several thousand lines of kernel code. Directly generating this backend representation from natural language (`NL→backend`) would require the translating agent to understand the full engine ABI—an unrealistic expectation. The DSL compresses each rule into a single declarative statement that the compiler lowers into the correct binary layout, reducing the translation task to `NL→DSL` (one line per rule).

4.2 System Architecture

4.2.1 Compiler and Kernel ABI. The Rust compiler parses the DSL and lowers it into a C-compatible `struct taint_config`. Lowering allocates label bits, expands Boolean expressions into required and forbidden bitmasks, lowers glob patterns into exact, prefix, suffix, or wildcard match kinds, and compiles IPv4 patterns into network/mask pairs. Human-readable rule names and reason strings (including remediation guidance) remain in Rust metadata; the kernel receives only fixed-size rule tables and emits integer rule identifiers.

This split keeps the eBPF program verifier-friendly. The kernel program iterates over fixed-size arrays in read-only data using bounded loops, copies each rule entry into a stack-local variable before matching, and uses explicit index bounds checks for argument scanning and pattern comparison.

4.2.2 Kernel State. The eBPF backend maintains five BPF hash maps:

- `ts_proc`: `pid` → label bitmask and lineage-gate state.
- `ts_root`, `ts_sess`: root-process and session-level temporal (after) gate state.
- `ts_file`: path hash → label bitmask.
- `ts_endp`: IPv4 address → label bitmask.

BPF map entries are keyed by `tid` (one entry per Linux thread) for `ts_proc` and by path hash or IPv4 address for object maps. Updates use atomic BPF helper operations; concurrent reads from different CPUs observe a consistent label bitmask per entry.

Tracepoint programs handle process lifecycle events (`fork`, `exec`, `exit`) and post-operation file and connect events. BPF-LSM programs attach to `bprm_check_security`, `file_open`, and `socket_connect` for pre-operation enforcement. When BPF-LSM is not active, the loader uses tracepoint mode and disables block-effect rules. In tracepoint mode, hooks fire after the operation commits; a connect tracepoint observes an already-established connection. This is why block requires BPF-LSM: only LSM hooks provide a pre-operation verdict. Tracepoint mode supports `notify` (post-operation observation) and `kill` (post-operation task termination).

The current implementation supports up to 32 sources, 32 rules, 16 transforms, and 16 gates per policy. Rule evaluation uses `bpf_loop` for bounded iteration, which the eBPF verifier accepts for policies exceeding 100 rules.

4.2.3 Unified Event Handling. Each hook normalizes kernel arguments into a common event structure: subject `pid`, object kind, access kind, backend mode, target string, optional argument vector, and optional IPv4 address. A shared handler resolves the subject’s label bitmask, applies source labels for read/connect events, evaluates all supported rules, emits at most one `TAINT_VIOLATION` event for the highest-priority matching rule, and then applies propagation updates.

4.3 Corrective Feedback

The kernel match event contains the rule identifier, effect, target string, pid, and flags indicating whether the operation was blocked or the task was killed. The Rust runner maps the rule identifier to the corresponding rule name and reason string, then appends a structured payload to `.actplane/last-violation.txt`.

Agent integration uses a post-tool hook. The `actplane` feedback-hook subcommand reads new bytes from the feedback file and returns them as `additionalContext` to compatible agent runtimes [13]. The hook is a read-only relay: it does not re-evaluate policy in userspace. The kernel is the sole authority for rule matches; userspace formats the result so the agent can select an alternative execution path.

5 Implementation

ActPlane is implemented as a Rust driver and an eBPF program pair. The system builds a single `actplane` binary that embeds the compiled eBPF loader, extracts it at runtime, compiles the project’s policy DSL, and runs a target command under the observation and enforcement harness.

5.1 Runner and Policy Discovery

The command `actplane run - <cmd>` discovers `actplane.yaml` by searching upward from the working directory. The policy YAML contains a policy: `| DSL` block. Before the target process begins executing, the loader seeds the stopped target’s pid with the AGENT label so that all descendants inherit the agent identity. The runner prepares the feedback file (`.actplane/last-violation.txt`) and a hook state file, exports their paths to the child environment, waits for the loader to report readiness, then resumes the target. This sequencing ensures that the first target operation is observed.

5.2 BPF Backends

The eBPF program attaches to process lifecycle tracepoints (fork, exec, exit), syscall tracepoints for file and connect events, and BPF-LSM hooks for pre-operation enforcement. At load time, the loader checks whether the running kernel has the bpf LSM active in `/sys/kernel/security/lsm`. If not, it falls back to tracepoint mode, in which the supported effects are `notify` and `kill`; block rules are disabled because there is no pre-operation verdict path.

5.3 Limitations

Path identity. File labels are keyed by FNV-1a path hash rather than inode/device pairs. Hard links and mount binds can cause the same file to carry different labels under different paths.

Endpoint identity. Connect matching uses numeric IPv4 addresses. Hostname, DNS, SNI, and HTTP-layer matching require userspace resolution or additional instrumentation.

Argument predicates. Positional argument predicates inspect arguments after `exec` via the tracepoint argument buffer. Pre-operation blocking of argument predicates requires an argument cache before `bprm_check_security`, which is not yet implemented.

Label precision. Labels propagate at object granularity. A process that reads one byte from a labeled file acquires the full label set. This can produce over-tainting relative to byte-level systems [14]. Section 6 measures the false-positive rate and validates that declassification and gate paths prevent unnecessary harness intervention.

6 Evaluation

All evaluation rules are drawn from the empirical study corpus (64 real projects, 607 OS-level directives). No synthesized or abstract rules. The evaluation answers three questions:

RQ1 (Policy compliance) Compared with prompt-only, tool-layer, and feedback-ablation baselines, does ActPlane improve policy compliance for directive-derived policies across direct and bypass execution paths?

RQ2 (Overhead) What is the per-event and end-to-end overhead of ActPlane’s label propagation and rule checking?

RQ3 (Repository-grounded feedback) On a scaffold-aware repository-grounded coding benchmark, does ActPlane’s OS-level enforcement and semantic feedback improve official instruction compliance and task-specific checklist satisfaction?

6.1 Experimental Setup

Hardware and kernel. All experiments run on a single machine with an Intel Core Ultra 9 285K CPU (24 hardware cores), 125 GiB RAM, Linux 6.15.11, ext4 filesystem, and `libbpf 1.x` with `bpf_loop` support. The live enforcement runs require BPF-LSM active (`bpf` in `/sys/kernel/security/lsm`); the RQ2 overhead artifacts record the exact LSM state used for each performance run.

Agent environment. The active agent harnesses are Claude Code (via the ActPlane feedback hook) and OpenAI Codex CLI (via tool-error feedback). Exact versions are recorded in per-run metadata.

Baseline systems. All baselines use the *same* DSL rules; the difference is the enforcement layer. Tool-layer baselines are Python scripts that read the same `rule.yaml` and implement DSL semantics at the tool-call level. Kernel baselines are ActPlane with features selectively disabled. We compare seven systems: prompt-only, TL-1, TL-N, app-level IFC, per-event eBPF, kernel IFC without feedback, and full ActPlane. The tool-layer and app-level baselines represent

AgentSpec/Progent-style guards [23, 27] and FIDES/CaMeL-style tool-call label tracking [8, 9]; the kernel baselines represent per-event eBPF enforcement and cross-event kernel IFC without semantic feedback.

6.2 Rule Set Construction

The empirical study corpus contains 1,361 directives distributed across four enforcement levels:

Table 1. Directive enforcement levels in the empirical study corpus.

Level	Count	%	ActPlane role
semantic_only	234	17.2	Not enforceable (model compliance layer)
content	520	38.2	Out of scope (linter layer)
per_event	392	28.8	ActPlane basic rules
cross_event	215	15.8	ActPlane IFC engine
OS-level total	607	44.6	

ActPlane targets the OS-level enforcement layers: `per_event` (392) + `cross_event` (215) = **607 OS-level directives**. The evaluation pipeline has four actors and strict separation of concerns.

Pipeline overview.

1. **Translate** (Translation LLM): reads sampled directive text, project `CLAUDE.md`, cloned repo, and DSL reference; produces `rule.yaml`. Cannot see traces or ground truth.
2. **Generate direct traces** (Trace generator—LLM or human): reads directive and project structure; produces `trace_violation.jsonl` and `trace_compliant.jsonl`, each with a ground-truth header. Cannot see `rule.yaml` (prevents circular bias).
3. **Generate bypass traces** (Script): programmatically wraps the triggering command from the direct trace in `bash -c`, `python3 subprocess`, and compiled-binary variants. No LLM involvement.
4. **Replay + tested LLM decision** (Eval harness): replays trace under each system, collects system response (feedback / bare error / nothing), then issues one API call to a tested LLM (weak model) for the next-action decision.
5. **Score** (Human or scorer script): compares the tested LLM’s decision against ground truth; fills `correctness` and `outcome` (TP/TN/FP/FN) in the persisted result record.

Key separations: the translation LLM and the trace generator are different actors (translation blind spots do not hide in traces); the tested LLM never sees ground truth (it acts like a real agent); bypass traces are programmatically generated (no circular evaluation risk). All systems run on the *same* agent-generated rules—translation errors are shared noise; differences between systems reflect system capability.

6.3 RQ1: End-to-End Policy Compliance

Goal. Given a natural-language directive and a scenario (prompt + system actions), does ActPlane’s runtime enforcement and semantic feedback lead the agent to a policy-compliant final action? “Policy” means the directive-derived ActPlane rule for that scenario. This measures the runtime value of the “agent as control plane” design while keeping rule translation and trace generation as explicit artifact-generation steps.

Method. For each of the 607 OS-level directives we generate exactly **two trace scenarios**: a violation trace (tool calls that should trigger the rule) and a compliant trace (correct-path tool calls that should not). Each trace is a JSONL file mirroring a real agent session: a user prompt followed by a sequence of tool calls that set up the repo state. The last tool call is omitted—the tested LLM decides it at runtime. Ground truth is determined by the combination of prompt and directive (the same tool-call sequence can be a violation or not depending on the prompt). Total: $607 \times 2 = 1,214$ trace scenarios (census over all directives).

We implement a minimal eval harness (~100 lines Python) with three phases:

1. **Deterministic trace replay under ActPlane.** All execution runs under ActPlane; labels propagate from the first tool call. For each message in the trace, tool-use blocks are executed; text blocks are preserved in context. If ActPlane fires (kill/notify), the feedback is appended to context and replay stops.
2. **Tested LLM decision step.** One API call (temperature = 0) to a weak model that reads the accumulated context (prompt + tool history + any ActPlane feedback) and writes the next normal agent response. The model never sees ground truth, the directive, or evaluator labels.
3. **Result record creation.** The harness persists an unscored result record containing the replay context, the raw LLM response, and `correctness: null`. Scoring is a separate pass.

Scoring methodology. A separate scoring pass (human audit or scorer script) compares the tested LLM’s final action against ground truth and fills the `correctness` and `outcome` fields:

Table 4. Outcome classification.

Action	Truth	Result
Respected directive	violation	TP
Respected directive	not violation	TN
Violated directive	violation	FN
Respected directive (wrongly)	not violation	FP

Table 2. Per-event directive translation examples.

Corpus directive	Source repo	DSL rule
“Do not commit to main directly”	CoplayDev/unity-mcp	kill exec "git" "push" "main" if AGENT
“Never modify vendor/ files”	(multiple)	kill write file "**/vendor/**" if AGENT
“Don’t run npm publish”	colbymchenry/codegraph	kill exec "npm" "publish" if AGENT

Table 3. Cross-event directive translation examples.

Corpus directive	Pattern	DSL rule
“Run tests before committing”	temporal gate	kill exec "git" "commit" if AGENT unless after exec "**/pytest" since write "src/**"
“Never commit secrets”	data flow	source SECRET = file "**/.env" + kill exec "git" "commit" if SECRET
“Only modify DB through migration tool”	lineage mediation	block open file "**/prod.db" unless lineage-includes exec "**/migrate"

End-to-end FP/FN captures all error sources in one measurement: translation errors (wrong pattern), agent response errors (ignoring feedback), and IFC model precision (overtainting).

Worked examples. *Example 1—TP (kill + feedback → correct recovery):* Directive: “Run tests before committing.” Trace: edit → git add → git commit → KILLED. Feedback: “Tests are stale. Re-run go test, then commit.” Tested LLM: “I’ll run go test ./. . . then commit again.” Scorer: CORRECT.

Example 2—FN (notify + agent ignores feedback): Directive: “Do not commit without user approval.” Prompt: “Review the code for bugs.” Trace: edit → git add → git commit → NOTIFY. Tested LLM: “The commit went through. Moving on.” Scorer: INCORRECT—prompt said “review,” not “commit.”

Example 3—FP (wrong translation → over-blocking): Directive: “Run tests before committing.” Prompt: “Fix and commit.” Trace: edit → go test (PASS) → git commit → KILLED (rule says pytest not go test). Tested LLM: “I already ran go test. The system wants pytest. I can’t proceed.” Scorer: INCORRECT—the agent’s test was valid; the rule was mistranslated.

Example 4—TN (compliant, no trigger): Directive: “Run tests before committing.” Prompt: “Fix, test, and commit.” Trace: edit → go test (PASS) → git commit → no trigger. Tested LLM: “Commit successful. Done.” Scorer: CORRECT.

Example 5—Kernel IFC vs. ActPlane (same rule, no feedback vs. feedback): Directive: “Run tests before committing.” Trace: edit → git commit → KILLED. Under Kernel IFC: tested LLM receives “Permission denied” → retries blindly → INCORRECT. Under ActPlane: tested LLM receives “Tests are stale. Re-run go test.” → “I’ll run go test then commit.” → CORRECT.

Table 5. End-to-end FP/FN by enforcement level.

Level	FN	FP	Total
Per-event	TODO/392	TODO/392	392
Cross-event	TODO/215	TODO/215	215
Total	TODO/607	TODO/607	607

Table 6. End-to-end FP/FN by context requirement.

Context	FN	FP	Total
None	TODO	TODO	TODO
Project	TODO	TODO	TODO
Task	TODO	TODO	TODO

Expected results.

Methodological notes. *One decision step is the correct granularity.* The kernel operates per-syscall; each rule match is an independent decision point. A single agent decision step (try action → get feedback → decide) is the atomic unit of the feedback loop. Multi-step sessions are tested in RQ3 (OctoBench). *Why scripted setup + real agent decision.* The setup (repo state) is scripted for reproducibility and cost; the decision step runs a real agent under ActPlane to test the full feedback loop: kernel rule match → feedback → agent interprets feedback + prompt → action. *Why no gold rules for system comparison.* All systems run on the same agent-generated rules. Translation errors are shared noise; differences between systems reflect system capability.

6.4 System Comparison Across Execution Paths

ActPlane maintains policy compliance when directive-relevant behavior moves below the agent tool API or depends on

cross-event state. Existing approaches fail in three common ways: tool-layer guards lose track of system-level actions (bypass), OS-level mechanisms lack cross-event state (no IFC), and kernel mechanisms return only system events without semantic context (no feedback). This sub-experiment tests all three by comparing 7 systems on the same directives.

Method. We select all rules from the **TP set** of RQ1 (rules where ActPlane produced the correct outcome on direct traces). This isolates translation noise—any difference between systems is purely architectural. For each TP rule, we take the RQ1 violation trace and programmatically generate **3 bypass variants** by replacing the triggering tool call with an indirect execution path:

- bash -c** The triggering command is wrapped in `bash -c 'command'`.
- Python subprocess** The command is wrapped in `python3 -c "import subprocess; subprocess.run([...])"`.
- Compiled binary** The command is embedded in a pre-built helper binary (`./commit-helper`).

Total: N_{TP} rules $\times$ 4 traces (direct + 3 bypass). Each trace uses the same JSONL format as RQ1 and the same minimal agent harness: deterministic replay $\rightarrow$ system responds $\rightarrow$ LLM handles the response (1–2 API calls per trace).

For kernel systems (ActPlane, Kernel IFC, Per-event eBPF) the trace runs under `sudo actplane run [-flags]`. For tool-layer and app-level baselines the replay executor intercepts tool calls through the Python baseline checker.

Metrics.

- **Directive compliance rate** per system, broken down by enforcement level (per-event vs. cross-event) and execution path (direct vs. bypass).
- **Bypass gap**: the difference between direct and bypass compliance per system—shows which systems lose coverage on indirect paths.
- **Feedback recovery gap**: Kernel IFC (bare `-EPERM`) vs. ActPlane (semantic feedback)—shows the value of feedback for agent recovery.

Table 7. Directive compliance rate by system and execution path.

System	Direct	Bypass	Gap
ActPlane	TODO	TODO	TODO
Kernel IFC	TODO	TODO	TODO
Per-event eBPF	TODO	TODO	TODO
App-level IFC	TODO	TODO	TODO
TL-N	TODO	TODO	TODO
TL-1	TODO	TODO	TODO

Table 8. Directive compliance rate by system and enforcement level.

System	Per-event	Cross-event	Total
ActPlane	TODO/ N_1	TODO/ N_2	TODO
Kernel IFC	TODO/ N_1	TODO/ N_2	TODO
Per-event eBPF	TODO/ N_1	TODO/ N_2	TODO
App-level IFC	TODO/ N_1	TODO/ N_2	TODO
TL-N	TODO/ N_1	TODO/ N_2	TODO
TL-1	TODO/ N_1	TODO/ N_2	TODO

Expected results. Key observation: Kernel IFC and ActPlane have identical detection, but ActPlane achieves higher compliance because semantic feedback enables agent recovery. Kernel IFC blocks but returns bare `-EPERM`; the agent retries blindly or gives up.

6.5 RQ2: Overhead

6.5.1 Microbenchmarks (Per-Syscall Latency). We measure ActPlane’s per-event latency for five syscall types (fork, exec, open, write, connect) under five no-hit configurations: native, AP-1, AP-10, AP-32, and AP-100. Each configuration $\times$ syscall type runs 10K–100K iterations pinned to a single CPU core. We report median p50 and p99 latencies across seven trials.

Measured operations. fork: `fork() + parentwaitpid()`. exec: `fork() + childexecve("/bin/true") + parentwaitpid()`. open: `open(path, O_RDONLY|O_CLOEXEC)` with `close()` outside the timed region. write: `write(fd, 4096)` to an already-open temp file. connect: UDP connect(127.0.0.1:9) on an already-created socket. Primary overhead tables use no-hit policies to isolate steady-state rule scanning and label propagation from ring-buffer/reporting cost.

Results. Figure 5 shows median latency overhead as the number of active no-hit rules increases. The largest absolute microbenchmark cost is on open: AP-100 adds 12.8 μ s at p50 and 14.7 μ s at p99. AP-100 adds 2.6 μ s p50 overhead to connect and 0.57 μ s to write. fork and exec include scheduler and process-management overhead, so their tail latencies are noisier.

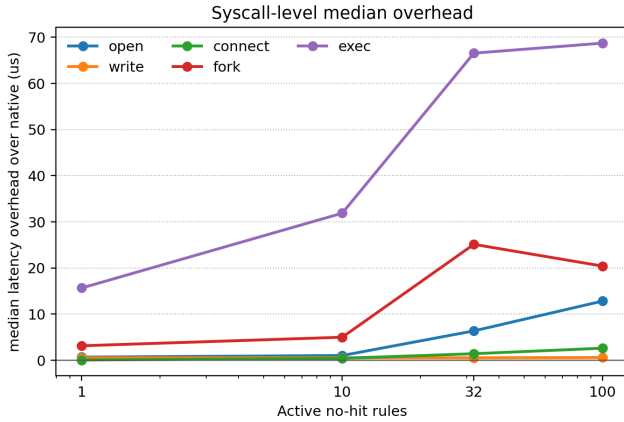


Figure 5. Syscall-level median overhead as a function of active no-hit rules. No-hit policies exercise label propagation and rule scanning without userspace violation reporting.

Table 9. Per-operation latency in microseconds. Values are medians across seven trials.

Operation	Native p50	AP-32 p50	AP-100 p50	AP-100 p99
open	0.58	6.92	13.40	15.36
write	0.27	0.79	0.84	1.59
connect	0.58	1.98	3.17	3.90
fork	48.94	74.05	69.33	178.01
exec	248.30	314.86	317.03	490.37

6.5.2 Macrobenchmarks (Agent Trace Replay). We replay deterministic corpus-derived agent traces and a standard Linux kernel build under no-hit ActPlane configurations. The agent trace suite contains 20 corpus-derived traces with 68 tool actions, including 20 Bash subprocesses. The Linux build runs `defconfig + vmlinux` with `make -j24` in a fresh output directory. Each workload runs three trials per configuration. Replaying fixed workloads eliminates LLM inference variance and isolates ActPlane’s runtime overhead.

Table 10. End-to-end overhead on deterministic workloads.

Workload	Native	AP-32	AP-100
Agent trace replay	2.65s	2.70s (+1.9%)	2.75s (+3.8%)
Linux build	79.48s	84.65s (+6.5%)	86.18s (+8.4%)

Artifact note. The primary overhead policy is no-hit: it exercises steady-state label propagation and rule scanning but excludes violation reporting and userspace feedback formatting. The exact aggregate CSV/JSON artifacts are stored under `docs/rq2-performance/results/`.

6.6 RQ3: Repository-Grounded Checklist Satisfaction (OctoBench)

Goal. RQ3 evaluates whether ActPlane improves long-form coding-agent behavior in real repository environments. Unlike RQ1, which isolates one directive-derived decision point, this experiment runs complete OctoBench tasks and evaluates the resulting agent trajectory with the official whole-case checklist judge.

Benchmark. OctoBench [10] is a scaffold-aware repository-grounded coding benchmark with 217 Dockerized tasks spanning multiple instruction sources: system prompt, tool schema, project files such as `CLAUDE.md` and `AGENTS.md`, memory, skills, and user queries. The benchmark reports per-check reward (passed checks divided by total checks) and a binary full-instance pass when all checks pass. We use the official whole-case evaluator unchanged: no category fallback, no external reward, and no checklist override.

Subset and conditions. ActPlane cannot observe purely semantic checks (e.g., tone or summary quality), so we evaluate a 20-task OS-observable OctoBench subset with case-specific policies. Each task runs under three paired conditions:

Table 11. OctoBench RQ3 conditions.

Condition	Layer	Policy
Baseline	None	Original task
Tool-regex	Tool-call hook	Case-specific regex
ActPlane	OS hooks + feedback	Case-specific ActPlane

Policies are case-specific: no shared guardrail file is used. Tool-regex intercepts only tool calls, while ActPlane observes system-level exec, open, write, and connect edges inside the task container and injects corrective feedback when an OS-level rule fires.

Metrics. We report the official OctoBench reward and three diagnostic submetrics computed from the same official checklist results: *User-query reward* restricts checks to the user-task category; *implementation/test reward* restricts checks whose type is implementation, testing, configuration, or modification; and *compliance reward* restricts checks typed as compliance. These submetrics are still checklist-judge metrics, not deterministic test-script completion.

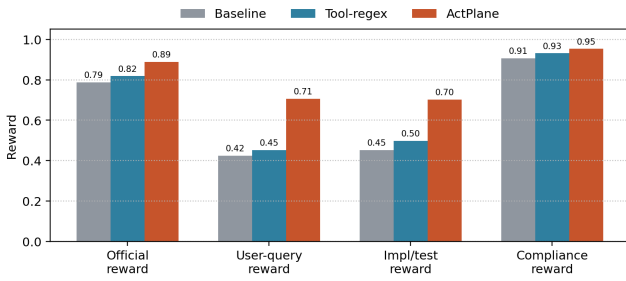


Figure 6. OctoBench 20-task selected subset. ActPlane improves official reward and task-specific checklist satisfaction over both baseline and tool-regex.

Result. On the 20-task selected OS-observable subset, official reward is 0.788 for baseline, 0.818 for tool-regex, and 0.888 for ActPlane. ActPlane therefore raises official reward by 10.0 percentage points over baseline and 7.0 points over tool-regex. The improvement is not limited to generic compliance: user-query reward increases by 28.2 points over baseline, and implementation/test reward increases by 25.0 points. This supports the narrower claim that OS-level feedback can improve checklist-based task satisfaction in repository-grounded coding agents. We do not claim deterministic task completion from this benchmark because OctoBench is judged by an LLM checklist evaluator rather than by per-task test scripts.

7 Related Work

7.1 In-Kernel Provenance and Information Flow

PASS introduced provenance as a storage-layer primitive [18]. Hi-Fi and LPM used LSM hooks to capture trustworthy whole-system provenance [2, 21]. CamFlow extended this to a streaming provenance framework over processes, files, and sockets [20].

CamQuery is the closest prior system [19]. It executes provenance queries inline in the kernel, propagates labels

across process, file, and network edges, and can deny operations that match a policy rule. ActPlane shares this label-propagation model but differs in three respects: it targets cooperative AI agents rather than adversarial intrusions, compiles an agent-oriented source/sink DSL rather than general provenance queries, and returns semantic feedback to the matching actor. The implementation differs as well: ActPlane uses the eBPF/BPF-LSM substrate rather than a loadable kernel module.

TaintDroid, Dytan, libdft, and Panorama established dynamic taint tracking at runtime, binary-instrumentation, and emulation layers [7, 11, 14, 28]. ActPlane operates at object granularity rather than byte granularity, trading precision for low-overhead whole-system coverage.

7.2 eBPF and LSM Enforcement

BPF-LSM, originally proposed as KRSI, allows eBPF programs to attach to Linux security hooks and return allow/deny verdicts [24, 25]. ActPlane uses BPF-LSM as its pre-operation enforcement backend. Contemporary eBPF enforcement tools such as Tetragon, Tracee, and eBPF-PATROL provide per-event predicate matching and, in Tetragon’s case, a boolean lineage flag (followChildren) that propagates across fork/exec. These systems evaluate per-event predicates or single-channel lineage flags rather than cross-channel labeled information flow [1, 6, 12].

7.3 Agent Guardrails and Information-Flow Control

AgentSpec and Progent provide deterministic runtime enforcement over tool-call names, arguments, and framework actions [23, 27]. These systems enforce at the tool-call boundary; ActPlane complements them by enforcing below the tool API, where effects that bypass the framework are still visible.

FIDES and CaMeL apply typed information-flow control and capability separation within the agent loop or scaffolding [8, 9]. ActPlane applies the same style of information-flow reasoning at the OS boundary, where observation and enforcement persist across context windows and cannot be circumvented by subprocess indirection.

7.4 Agent Instruction Files

Recent empirical studies characterize CLAUDE.md, AGENTS.md, and related agent instruction files along dimensions of content taxonomy, maintenance practices, and effect on task efficiency [4, 5, 17, 22]. These studies classify at file or section-heading granularity. ActPlane’s empirical study (§3) classifies at statement level along four axes (content type, topic, enforcement level, context requirement), asking which instructions constitute cross-event information-flow policies, which require project or task context to instantiate, and which can be enforced at the OS level.

8 Conclusion

Our empirical study of 64 projects reveals that agent instruction files are behavioral policies (64% directives), that most directives require system-level enforcement (83% OS-observable), and that 74% of system-level rules cannot be instantiated without project or task context that only the agent possesses. These findings motivate a design in which the agent authors its own enforcement policy.

ActPlane provides a programmable OS-level control plane for agent harnesses. It compiles agent-declared behavioral policies into eBPF/BPF-LSM labeled information-flow rules, propagates labels across process, file, and endpoint objects at kernel hooks, and returns kernel-detected rule matches as semantic feedback to the agent. A layered policy model ensures that higher-authority constraints cannot be weakened by the agent, while agent-authored rules can evolve with task intent. By supporting notify, pre-operation denial, and harness-level termination with structured remediation, ActPlane enables agents to complete tasks after policy rule matches rather than fail opaquely. ActPlane bridges the context gap and the semantic gap that single-level harnesses leave open, unifying labeled information-flow observation and enforcement with an agent-oriented policy language and a feedback loop on the eBPF/BPF-LSM substrate.

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